

# Julian C. Stamm

M.Sc. Computer Science · Real-Time GI & Neural Rendering

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## EDUCATION

- Oct 2025 – present    **University of Bonn**, M.Sc. in Computer Science
- Focus on Real-Time GI, Neural Rendering, and Machine Learning
- Oct 2021 – Sept 2025    **University of Bonn**, B.Sc. in Computer Science
- Grade: 1.1 ([top 3%](#) 📄)
  - Bachelor's Thesis (1.0): Bidirectional Neural Radiance Caching [[full-res](#) 📄] [[low-res](#) 📄]

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## PUBLICATIONS

- May 2026    [Photon-Driven Neural Radiance Caching](#) 📄
- Introduces a new photon-based training method for Neural Radiance Caching, reducing FLIP error by up to two-thirds in caustic scenes with only a ~2 ms frame-time overhead. A continuation of my Bachelor's thesis, presented at I3D 2026.
- Julian C. Stamm, Tom Kneiphof, Reinhard Klein  
[10.1145/3807895.3807923](#) 📄 (I3D Companion 2026)

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## EXPERIENCE

- Feb 2026 – Mar 2026    **Teaching Assistant for "Foundations of 4D/6D Object Capture for Virtual Environments"**, University of Bonn – Bonn, Germany
- Held daily block tutorials on NeRFs, IK, 3DGS, differentiable rendering, etc.
- Apr 2024 – present    **Teaching Assistant for "Introduction to Computer Graphics"**, University of Bonn – Bonn, Germany
- Held weekly tutorials on rasterization, ray tracing, BSDFs, splines, etc.
  - Assisted in creating exercises and exams
  - Developed and maintained a modern OpenGL/C++ framework used in the exercises and the animation competition
- May 2021 – Sept 2021    **Temporary Worker in Software Development Team**, Atlas Spare Parts GmbH – Ganderkesee, Germany
- Developed custom CAN bus logger.

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## PROJECTS

- May 2026 – present    [tsnn](#) 📄 (unpublished)
- High-performance Slang-native tiny-cuda-nn alternative using Slang's auto-differentiation.
- Implemented neural spline flows, hash grid encodings, MLPs, Adam, and more
  - Achieves full kernel fusion with Falcor path tracer
  - Slightly outperforms tiny-cuda-nn in my NRC benchmark
  - Written for my current lab research on real-time neural importance sampling
- Feb 2025 – present    [PhotonNRC](#) 📄
- Falcor implementation of the poster "Photon-Driven Neural Radiance Caching".
- Rebuilt the method as a clean Falcor render graph, improving modularity and performance
  - Implemented SPPM with hardware-accelerated inverse radius search and stochastic accumulation, NRC via tiny-cuda-nn, and SPPC training

Feb 2025 – Sept 2025 [BiNRC](#) ↗

- OptiX implementation of my Bachelor's thesis "Bidirectional Neural Radiance Caching".
- Built an unbiased path tracer with MIS and NEE and a basic Disney principled BSDF
  - Implemented SPPM with hardware-accelerated inverse radius search
  - Implemented NRC via tiny-cuda-nn with bidirectional and photon-driven (Photon-NRC) training

Aug 2024 – present [nBounce](#) ↗

- Personal real-time path tracer with software BVH construction and traversal.
- Written in Rust and wgpu
  - SAH-based BVH construction, Owen-scrambled RQMC sampling, VNDF importance sampling

Oct 2023 – Mar 2024 [Stable Real-Time Simulation of Mesoscopic Glints](#) ↗

- Computer Graphics Lab, part of my B.Sc. in Computer Science.
- Implemented the discrete stochastic microfacet model from "Real-Time Rendering of Glinty Appearances Using Distributed Binomial Laws on Anisotropic Grids" by Thomas Deliot and Laurent Belcour

Aug 2023 – present [gltemplate](#) ↗

- OpenGL 4.6 framework developed for CG exercises.
- Written in C++ and CMake, RAII-based resource management, texture/mesh loading utilities

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## OTHERS

Apr 2026 – present [bevy\\_trail](#) ↗

Work-in-progress bevy plugin for high-performance GPU-based trails in wgpu.

Sept 2025 – present **P2P Multiplayer Game**

Working on a Python-moddable P2P-multiplayer card game in Rust using bevy.

June 2023 – Sept 2023 [monarch-tree](#) ↗

- Procedural tree and butterfly-swarm animation for the course "Introduction to Computer Graphics"
- Growing the tree with a self-devised model that steers branches using boids-like forces (up, spread, forward, noise).

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## SKILLS

**Languages:** C++, CMake, GLSL, CUDA, Slang, Python, Rust, WGSL, Java

**APIs:** OpenGL, OptiX, Falcor, wgpu, bevy, Godot, PyTorch

**Research Interests:** Real-Time GI, Neural Rendering, Path Tracing, Photon Mapping, Machine Learning, Normalizing Flows

**Spoken Languages:** German (native), English (TOEFL iBT: 110/120)